A uniform closed form for modular Schur numbers $S_m(k, \ell)$

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Abstract

The modular Schur number $S_m(k, \ell)$ of Chappelon–Marchena–Domínguez is the greatest N such that $[1, N]$ admits a k-partition into classes that are ℓ-sum-free modulo m. We record the uniform closed form $S_m(k, \ell) = m/\gcd(m, \ell - 1) - 1$ for every $m \geq 2$, $\ell \geq 2$, and $k \geq m/\gcd(m, \ell - 1) - 1$. The formula is a two-paragraph corollary of the singleton criterion (Theorem 4) and Corollary 3 of D'orville–Sim–Wong–Ho (2025); it extends their Corollary 5 (the coprime case $\gcd(m, \ell - 1) = 1$) to all values of the gcd, and settles the stable-k regime of their Problem 1, part 5 for all m. The main theorem is formalized in Lean 4 with axiom dependencies only `propext`, `Classical.choice`, `Quot.sound`. Beyond the closed form, we give a sharp k_0 for prime moduli via an iterated-sumset saturation lemma, and record a small composite counterexample ($C = \{1, 5\} \subseteq \mathbb{Z}/12$) that refutes the natural "only singletons are safe" extrapolation.

1 Introduction

The classical Schur number $S(k)$ is the largest N such that $\{1, \ldots, N\}$ admits a k-partition into sum-free sets. Values are known only through $S(5) = 160$ [Heule 2017]; $S(6)$ is the content of Erdős problem 483 and remains open. Generalizations to longer equations were studied by Beutelspacher and Brestovansky [Beutelspacher–Brestovansky 1982] and are now textbook material [Landman–Robertson 2014].

Chappelon, Marchena, and Domínguez [Chappelon et al. 2013] introduced the modular version of the generalized Schur number.

Definition 1.1. A set $S \subseteq \mathbb{Z}$ is *ℓ-sum-free modulo m* if no $x_1, \ldots, x_\ell \in S$ (repetitions allowed) and $y \in S$ satisfy $x_1 + \cdots + x_\ell \equiv y \pmod{m}$. The *modular Schur number* $S_m(k, \ell)$ is the greatest $N \geq 0$ such that $[1, N]$ admits a k-partition into ℓ-sum-free sets modulo m.

Chappelon et al. [Chappelon et al. 2013] closed $m \in \{1, 2, 3\}$ and observed that the determination of $S_m(k, \ell)$ "seems to be much more difficult for moduli $m \geq 4$" [Chappelon et al. 2013, §6]. D'orville, Sim, Wong, and Ho [D'orville et al. 2025] subsequently closed $m \in \{4, 5, 6, 7\}$ via residue-by-residue case analysis on $\ell \bmod m$, leaving $m \geq 8$ as Problem 1, part 5 of their open problems. The pattern visible across all ten tables (once k is large enough, $S_m(k, \ell)$ is a function of $\gcd(m, \ell - 1)$ alone) was not formulated as a unified statement.

Writing $d := \gcd(m, \ell - 1)$ and $n := m/d$, the following theorem makes that pattern explicit, with no hypothesis on m and no case split on $\ell \bmod m$.

Theorem 1.2. *For every $m \geq 2$, $\ell \geq 2$, and $k \geq n-1$,*

$$S_m(k,\ell) = n - 1 = \frac{m}{\gcd(m, \ell - 1)} - 1.$$

The proof is elementary: both bounds reduce to the observation that n itself is a singleton that fails ℓ-sum-freeness. We give it in the next two sections. Theorem 1.2 settles the stable-k regime ($k \geq n-1$) of [D'orville et al. 2025, Problem 1, part 5] for all $m \geq 2$.

The formula follows from [D'orville et al. 2025, Theorem 4 + Corollary 3] by choosing the optimal singleton: the smallest $a \in [1, m-1]$ for which Corollary 3's hypothesis $\ell \equiv 1 \pmod{m/\gcd(a,m)}$ holds is $a = n$, giving the upper bound $S_m(k,\ell) < n$. [D'orville et al. 2025] used Corollary 3 only through Corollary 5 (the coprime case $\gcd(m, \ell-1) = 1$, where the hypothesis holds for no $a \leq m-1$); they did not record the optimization for general gcd. The genuinely new content of the present note is the threshold k_0 analysis in Section 5: a cyclic-group iterated-sumset saturation lemma (Theorem 5.3), a sharp k_0 for prime moduli, and the explicit witness $\{1,5\} \subseteq \mathbb{Z}/12$ showing the prime case does not generalize naively.

1.1 Formalization

The proof is short but has a form that is easy to misstate (we previously confused the role of n versus m in the forbidden tuple). Theorem 1.2 has been formalized in Lean 4 to remove any doubt; the core closed-form proof was completed by Aristotle [Achim et al. 2025]. The Lean development defines $S_m(k,\ell)$ both in its original integer form (Definition 1.1) as `schurMod` and in its residue-level form as `schurModResidue`; Lemma 2.1 below is formalized as the equivalence `schurMod_eq_schurModResidue`, so the machine-verified theorem

```
theorem schurMod_eq (m k l : N) (hm : 2 <= m) (hl : 2 <= l)
    (hk : m / Nat.gcd m (l - 1) - 1 <= k) :
    schurMod m k l = m / Nat.gcd m (l - 1) - 1
```

states Theorem 1.2 verbatim and compiles with axiom dependencies only `propext`, `Classical.choice`, `Quot.sound`. Source is at https://github.com/mysticflounder/modular-schur.

1.2 Organization

Section 2 establishes base lemmas. Section 3 proves the upper bound $S_m(k,\ell) \leq n-1$. Section 4 proves the matching lower bound. Section 5 analyzes the threshold $k_0(m,\ell)$ and records a tight prime-modulus result. Section 6 lists corollaries for specific m. Section 7 poses open questions on tight k_0 for composite m.

2 Preliminaries

Throughout, $m \geq 2$ and $\ell \geq 2$ are fixed integers, $d := \gcd(m, \ell-1)$, and $n := m/d$. Note $dn = m$ exactly, and $d \mid (\ell - 1)$ by definition.

Lemma 2.1 (Residue reduction)**.** *Let $N \leq m-1$. The integers $1, 2, \ldots, N$ have pairwise distinct nonzero residues modulo m, and a k-partition of $[1,N] \subseteq \mathbb{Z}$ into classes $C_1 \sqcup \cdots \sqcup C_k$ is valid (each C_i is ℓ-sum-free modulo m in the sense of Definition 1.1) if and only if the corresponding k-partition of the residue set $\{1,\ldots,N\} \subseteq \mathbb{Z}/m$ into $\bar{C}_1 \sqcup \cdots \sqcup \bar{C}_k$ has each $\bar{C}_i$ ℓ-sum-free modulo m. Consequently $S_m(k,\ell)$, as defined on integers in Definition 1.1, equals its residue-level counterpart.*

Proof. All elements of $[1, N]$ lie in $[1, m-1]$, and $[1, m-1] \hookrightarrow \mathbb{Z}/m$ is injective (if $a \equiv b \pmod{m}$ with $1 \le a, b \le m-1$ then $|a-b| < m$ forces $a = b$). Under this injection each C_i corresponds bijectively to a subset $\bar{C}_i \subseteq \{1, \ldots, m-1\} \subseteq \mathbb{Z}/m$. For any $(x_1, \ldots, x_\ell; y) \in C_i^\ell \times C_i$, the congruence $x_1 + \cdots + x_\ell \equiv y \pmod{m}$ in $\mathbb{Z}$ is equivalent to $\bar{x}_1 + \cdots + \bar{x}_\ell = \bar{y}$ in $\mathbb{Z}/m$; conversely every ℓ-tuple in $\bar{C}_i$ lifts uniquely to C_i. Thus ℓ-sum-freeness transfers in both directions, and the two definitions of $S_m(k, \ell)$ agree. (This equivalence is formalized in Lean as `schurMod_eq_schurModResidue`.) □

Lemma 2.2 (Universal upper bound, Chappelon et al. 2013)**.** *For every $\ell \ge 1$ and every $k \ge 1$, $S_m(k, \ell) \le m - 1$.*

Proof. If $N \ge m$, then $m \in [1, N]$ lies in some class C. The tuple $(m, \ldots, m; m) \in C^{\ell+1}$ satisfies $m \cdot \ell \equiv 0 \equiv m \pmod{m}$, violating ℓ-sum-freeness. (This is [Chappelon et al. 2013, Eq. (2)]; formalized in Lean as `no_valid_partition_of_ge_m`, and together with Lemma 2.1 justifies the $N \le m - 1$ search cap used in the Lean definition via `schurMod_is_greatest`.) □

Lemma 2.3 (Singleton safety)**.** *For $r \in \{1, \ldots, m-1\}$, the singleton $\{r\} \subseteq \mathbb{Z}/m$ is ℓ-sum-free modulo m if and only if $(\ell - 1)r \not\equiv 0 \pmod{m}$; equivalently, if and only if $n \nmid r$.*

This is [D'orville et al. 2025, Theorem 4] rewritten in the variables natural for our application; we include the derivation for self-containment.

Proof. $\ell\{r\} \cap \{r\} \neq \emptyset$ iff $\ell r \equiv r \pmod{m}$ iff $(\ell - 1)r \equiv 0 \pmod{m}$. Write $\ell - 1 = du$ so $\gcd(u, n) = 1$; then $(\ell - 1)r \equiv 0 \pmod{dn}$ iff $ur \equiv 0 \pmod{n}$ iff $n \mid r$. □

Lemma 2.4 (Monotonicity in k)**.** *If $[1, N]$ admits a valid k_0-partition, then for every $k \ge k_0$ it admits a valid k-partition.*

Proof. Pad with $k - k_0$ empty classes; the empty set is vacuously ℓ-sum-free. □

3 Upper bound

Theorem 3.1. *For all $m \ge 2$, $\ell \ge 2$, and $k \ge 1$,*

$$S_m(k, \ell) \le n - 1.$$

Proof. Suppose $N \ge n$. Since $1 \le n \le m$, we have $n \in [1, N]$; let C be the class of a valid k-partition containing n. Write $\ell - 1 = du$ with $u \in \mathbb{Z}$, so

$$(\ell - 1) \cdot n = du \cdot n = u \cdot (dn) = u \cdot m \equiv 0 \pmod{m}.$$

Hence $\ell n \equiv n \pmod{m}$. The tuple $(n, \ldots, n; n) \in C^{\ell+1}$ thus witnesses ℓ-sum-freeness failure in C, a contradiction. □

Remark 3.2. Equivalently, this is [D'orville et al. 2025, Corollary 3] applied at $a = n$. That corollary states $S_m(k, \ell) < a$ whenever $\ell \equiv 1 \pmod{m/\gcd(a, m)}$, and the smallest $a \le m - 1$ satisfying the hypothesis is $a = n$: any smaller a has $\gcd(a, m) < n$, so $m/\gcd(a, m) > \gcd(m, \ell - 1)$, and the hypothesis $m/\gcd(a, m) \mid (\ell - 1)$ would contradict $\gcd(m, \ell - 1)$ being maximal.

Remark 3.3. The two obstructions in prior arguments are the same: "$N \geq m$ is blocked by m" (Lemma 2.2, the $d = 1$ case $n = m$) and "$r = m/d$ is an unsafe singleton" (Lemma 2.3 at $r = n$). Both reduce to: the single element n is unsafe as a singleton, so any class containing it is unsafe. Once $N \geq n$, some class contains n.

4 Lower bound

Theorem 4.1. *For all $m \geq 2$, $\ell \geq 2$, and $k \geq n - 1$,*

$$S_m(k, \ell) \geq n - 1.$$

Proof. If $n = 1$ the conclusion reads $S_m(k, \ell) \geq 0$, vacuously true. Assume $n \geq 2$. Partition $[1, n-1]$ into singletons $D_r := \{r\}$ for $r = 1, \ldots, n-1$. By Lemma 2.3 each D_r is ℓ-sum-free (since $1 \leq r < n$ forces $n \nmid r$). This is a valid $(n-1)$-partition; for $k \geq n - 1$ pad with empty classes by Lemma 2.4. □

Proof of Theorem 1.2. Combine Theorems 3.1 and 4.1. □

5 The threshold $k_0(m, \ell)$

Let $k_0(m, \ell)$ denote the least k such that $S_m(k', \ell) = n - 1$ for every $k' \geq k$.

Theorem 5.1. $k_0(m, \ell) \leq \max(1, n - 1)$.

Proof. Theorem 4.1 exhibits a valid $(n-1)$-partition achieving $N = n - 1$ when $n \geq 2$. When $n = 1$ the value is zero, achieved trivially at $k = 1$. □

Theorem 5.2 (Covering lower bound)**.** *Let $\sigma(m, \ell) := \max\{|C| : C \subseteq \{1, \ldots, n - 1\}, C$ is ℓ-sum-free mod $m\}$. Then $k_0(m, \ell) \geq \lceil (n-1)/\sigma(m, \ell) \rceil$.*

Proof. If $[1, n-1]$ admits a valid k-partition, summing class sizes gives $n - 1 \leq k \cdot \sigma(m, \ell)$. □

5.1 Tight threshold for prime modulus

Theorem 5.3 (Large-ℓ coset criterion)**.** *Let $C \subseteq \mathbb{Z}/m$ with $|C| \geq 2$, pick $a_0 \in C$, and let $H := \langle C - C \rangle = g\mathbb{Z}/m$ with $g := m/|H|$. For every $\ell \geq |H| - 1$,*

$$\ell C = \ell a_0 + H,$$

and consequently C is ℓ-sum-free modulo m if and only if $g \nmid (\ell - 1)a_0$.

Proof sketch. ($\subseteq$) Every ℓ-sum $\sum c_i = \ell a_0 + \sum(c_i - a_0)$ lies in $\ell a_0 + H$ since $c_i - a_0 \in H$.

($\supseteq$) Let $D := C - a_0 \subseteq H$; then $0 \in D$ and $\langle D \rangle = \langle D - D \rangle = \langle C - C \rangle = H$. Since $0 \in D$, the iterated sumsets form an increasing chain $D \subseteq 2D \subseteq 3D \subseteq \cdots \subseteq H$, in which strict inclusions can occur at most $|H| - |D| \leq |H| - 1$ times; hence $kD = (k+1)D$ for some $k \leq |H| - 1$. Once $kD = kD + D$, the set kD is closed under adding any element of D, hence under adding any element of $\langle D \rangle = H$, so kD is a union of H-cosets; since $0 \in kD \subseteq H$, we get $kD = H$. For every $\ell \geq k$ (in particular $\ell \geq |H| - 1$), $\ell D = kD + (\ell - k)D = H + (\ell - k)D = H$. Therefore $\ell C = \ell a_0 + \ell D = \ell a_0 + H$.

The safety criterion follows: since $C \subseteq a_0 + H$, we have $\ell C \cap C \neq \emptyset$ iff $\ell a_0 + H = a_0 + H$ iff $(\ell - 1)a_0 \in H$ iff $g \mid (\ell - 1)a_0$. $\square$

Remark 5.4. The ($\supseteq$) direction is a cyclic-group version of the Cauchy–Davenport theorem [Cauchy 1813, Davenport 1935]: over $\mathbb{Z}/p$ prime, iterating $|A + B| \geq \min(p, |A| + |B| - 1)$ yields $|\ell C| \geq \min(p, \ell(|C| - 1) + 1)$, so any non-degenerate C ($|C| \geq 2$) has $\ell C = \mathbb{Z}/p$ once $\ell \geq p - 1$. Theorem 5.3 is the saturation statement inside an arbitrary cyclic subgroup $H \leq \mathbb{Z}/m$, with the explicit $\ell \geq |H| - 1$ threshold.

Corollary 5.5 (Sharp k_0 for prime moduli)**.** *Let p be prime. For every $\ell \geq p - 1$ with $\ell \not\equiv 1 \pmod p$,*

$$k_0(p, \ell) = p - 1.$$

Proof. For any $|C| \geq 2$ in $\mathbb{Z}/p$, the subgroup $\langle C - C \rangle$ is either 0 (ruled out) or $\mathbb{Z}/p$ (since $\mathbb{Z}/p$ has no nontrivial subgroups). By Theorem 5.3 with $g = 1$, $\ell C = \mathbb{Z}/p$, so C is not ℓ-sum-free. Hence $\sigma(p, \ell) = 1$, and Theorem 5.2 gives $k_0 \geq n - 1 = p - 1$; Theorem 5.1 supplies the match. $\square$

5.2 Composite moduli: a counterexample to the natural conjecture

Corollary 5.5 might suggest: for every m and every large ℓ with $\gcd(m, \ell - 1) < m$, only singletons are ℓ-sum-free, so $k_0(m, \ell) = n - 1$. This is false once m is composite.

Proposition 5.6. *For $m = 12$ and every $\ell \equiv 11 \pmod{12}$ (so $d = 2$, $n = 6$), the set $C = \{1, 5\} \subseteq \{1, \ldots, 5\}$ is ℓ-sum-free modulo 12. Consequently $\sigma(12, \ell) \geq 2$ and $k_0(12, \ell) \leq 3 < n - 1$.*

Proof. $C - C = \{0, \pm 4\}$, so $\langle C - C \rangle = 4\mathbb{Z}/12 = \{0, 4, 8\}$, $|H| = 3$, $g = 4$. Theorem 5.3 applies once $\ell \geq 2$. With $a_0 = 1$, the criterion $4 \nmid (\ell - 1) \cdot 1$ reads $4 \nmid \ell - 1$, i.e. $\ell \not\equiv 1 \pmod 4$. For $\ell \equiv 11 \pmod{12}$ we have $\ell \equiv 3 \pmod 4$, so $4 \nmid \ell - 1$, and C is ℓ-sum-free, giving $\sigma(12, \ell) \geq 2$.

For the k_0 bound, consider the 3-partition $[1, 5] = \{1, 4\} \sqcup \{2, 5\} \sqcup \{3\}$. Each pair-class has $C - C = \{0, \pm 3\}$, hence $H = \langle 3 \rangle = \{0, 3, 6, 9\}$, $|H| = 4$, $g = 3$; Theorem 5.3 applies for $\ell \geq 3$ with safety criterion $3 \nmid (\ell - 1)a_0$. For $\ell \equiv 11 \pmod{12}$ we have $\ell - 1 \equiv 10 \pmod{12}$, and $3 \nmid 10 \cdot 1$ as well as $3 \nmid 10 \cdot 2$, so both $\{1, 4\}$ (with $a_0 = 1$) and $\{2, 5\}$ (with $a_0 = 2$) are ℓ-sum-free. The singleton $\{3\}$ is safe by Lemma 2.3: $(\ell - 1) \cdot 3 \equiv 30 \equiv 6 \not\equiv 0 \pmod{12}$. Hence $k_0(12, \ell) \leq 3$. $\square$

6 Corollaries: recovered and extended formulas

Specializing Theorem 1.2:

- $m = 8$**:** for $k \geq 7$, $S_8(k, \ell)$ equals $0, 1, 3, 7$ according to $\gcd(8, \ell - 1) \in \{8, 4, 2, 1\}$ (residue pattern $\ell \bmod 8$).
- $m = 9$**:** for $k \geq 8$, $S_9(k, \ell)$ equals $0, 2, 8$ according to $\gcd(9, \ell - 1) \in \{9, 3, 1\}$.
- $m = 11$**:** for $k \geq 10$, $S_{11}(k, \ell) = 0$ if $\ell \equiv 1 \pmod{11}$, else 10.
- **Prime-power** $m = p^e$**:** writing $v := \min(v_p(\ell - 1), e)$ where v_p denotes the p-adic valuation, for $k \geq p^{e-v} - 1$,

$$S_{p^e}(k, \ell) = p^{e-v} - 1.$$

All of these stabilized rows are consistent with (and for $m = 4, 5, 6, 7$ recover) the tables of [D'orville et al. 2025, §2–3] and for $m = 1, 2, 3$ the theorems of [Chappelon et al. 2013, §1–2].

7 What stays open

1. *Tight k_0 for composite m.* The covering bound (Theorem 5.2) matches the easy upper bound $n-1$ for primes but not in general. For $m = 8$, ℓ even: $\sigma(8, \ell) = 4$ (witness $\{1, 3, 5, 7\}$), so the covering bound gives only $k_0 \geq 2$, while the true value is $k_0 = 3$. A closed form for $k_0(m, \ell)$ at composite m is open.
2. *Boundary $k < n - 1$.* Theorem 1.2 only governs $k \geq n - 1$. The smaller-k regime—where classes must be large and ℓ-sum-free—is empirically eventually periodic in ℓ mod m for fixed k (verified through $m \leq 13$), but I have no formula.
3. *Optimizing over ℓ.* For fixed k and $m \to \infty$, what is $\max_\ell S_m(k, \ell)$? The pointwise value is Theorem 1.2; the sup over ℓ becomes a divisor-counting problem on m.

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8 References

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